

Addressing Data Leakage in UAV Cyberattack Detection: A Realistic Machine Learning Benchmark

Şerif Mert Çeker

*Department of Computer Engineering
Fenerbahçe University
Istanbul, Türkiye
serif.ceker@stu.fbu.edu.tr*

Güldal Akçiçek

*Department of Computer Engineering
Fenerbahçe University
Istanbul, Türkiye
guldal.akcicek@stu.fbu.edu.tr*

Sıla İlayda Dönmez

*Department of Computer Engineering
Fenerbahçe University
Istanbul, Türkiye
sila.donmez@stu.fbu.edu.tr*

Aziz Burak Kaçar

*Department of Computer Engineering
Fenerbahçe University
Istanbul, Türkiye
aziz.kacar@stu.fbu.edu.tr*

Abstract—As the operational scope of Unmanned Aerial Vehicles (UAVs) expands, the need for security against cyberattacks such as GPS Spoofing and Jamming increases. This study addresses the common issue of data leakage in UAV intrusion detection datasets, specifically analyzing the "UAV Attack Dataset". While many recent studies achieve near-perfect detection accuracy, our work demonstrates that such results often stem from reliance on features that are unrealistic to obtain in real-time scenarios, such as direct jamming indicators and artificially strong GPS correlation ratios. By employing a domain-knowledge-based methodology to eliminate data leakage-prone features, we present a realistic evaluation benchmark for UAV cybersecurity. Models including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost) are evaluated. The XGBoost classifier demonstrated robust performance, achieving a 97.35% accuracy and F1-score, establishing a dependable and practical baseline for real-world UAV cybersecurity environments.

Keywords—Data leakage, drone, GPS spoofing, jamming attacks, machine learning, UAV communication, UAV security, unmanned aerial vehicles

I. INTRODUCTION

UAV deployments have expanded into communication relay and surveillance roles, which exposes flight control and navigation systems to new attack surfaces [1], [2]. Cybersecurity attacks against UAVs—most notably through signal jamming and Global Positioning System (GPS) spoofing—can result in substantial disruption of operations, leading to possible loss of control, mission failure, and severe safety risks.

In response, there has been a significant rise in the application of data-driven methods and machine learning (ML) approaches to detect and classify anomalous or malicious behavior within UAV communication networks. However, a critical issue persists in many of the proposed Intrusion Detection Systems (IDS): the reliance on datasets that inadvertently suffer from data leakage. Data leakage occurs when a model is

trained on features that would not realistically be available or observable at the time of an actual attack in a live production environment.

Our goal was not to use deceptive features that artificially inflate scores. However, our developed method demonstrated that even without these deceptive features (using strictly clean data), a high success rate of 97.35% can still be achieved. Restricting the feature set to physically observable telemetry allowed us to measure what detection accuracy is realistically achievable, without relying on features that presuppose knowledge of an ongoing attack.

The remainder of this paper is organized as follows. Section II reviews related work on UAV cybersecurity and intrusion detection systems. Section III describes the UAV Attack Dataset and the pre-processing methodology, including the proposed data leakage removal procedure. Section IV presents the experimental setup and the evaluated algorithms. Section V reports the classification results and feature importance analysis. Finally, Section VI concludes the paper and outlines directions for future work.

II. RELATED WORK

A. UAV Cybersecurity and IDS Systems

The vulnerability of UAVs to GPS-based attacks has been increasingly highlighted in recent literature. Real-world cases of spoofing and jamming have emphasized how these threats can drastically manipulate a drone's navigation system, leading to unintended route deviations. Previous studies [1]–[3] highlight that traditional encryption-based defenses are insufficient at the wireless layer, establishing a clear need for anomaly-based intrusion detection models that utilize realistic flight data. Furthermore, recent works emphasize the importance of cyber-physical feature fusion to secure swarms of UAVs [4], while advanced federated learning techniques are being

proposed to distribute intrusion detection effectively without compromising raw telemetry data [5].

B. Studies Using the UAV Attack Dataset

The "UAV Attack Dataset" used in this study [6] contains real-world recordings of GPS spoofing and jamming collected during actual UAV flight operations, making it highly valuable over simulated alternatives. Recent work has introduced improved benchmark datasets to address this challenge [7]. Several recent works have applied advanced learning techniques to this dataset. For instance, Wisanwanichthan and Thammawichai [8] proposed a lightweight IDS using knowledge distillation on the UAV Attack dataset, achieving a 98.59% F1-score. Other studies utilizing Convolutional Neural Networks (CNNs), Autoencoders (AE), and ensemble methods like XGBoost routinely report accuracies exceeding 99% when tackling UAV spoofing detection [9], [10].

C. Problem Definition and Gap

While previous studies establish strong theoretical architectures for UAV IDS, achieving 99% or even 100% accuracy often points to a fundamental flaw in the experimental setup: data leakage. Features that inherently summarize the attack state, such as raw signal quality disruption flags or extreme location disparities derived directly post-attack, allow models to effectively "cheat". In realistic flight conditions where early detection is vital, such attack-indicating features are misleading. This study fills this gap by purposely restricting ML models to a realistic feature space, sacrificing artificially inflated scores to demonstrate the true predictive capability of standard classification algorithms on authentic flight dynamics and control system states.

III. DATASET AND PRE-PROCESSING METHODOLOGY

A. UAV Attack Dataset Features

The dataset includes recordings of normal flights, jamming attacks, and GPS spoofing attacks. The initial high-dimensional raw flight logs (*uLog*) include categories such as Control & System Status, Navigation & Position, GPS Quality & EKF, Signal Health, and Kinematics & Velocity.

B. Data Leakage and Feature Selection

To ensure a realistic evaluation environment, a domain-driven methodology was utilized to eliminate features prone to data leakage.

- 1) **Explicitly Leakage-Prone and Zero-Variance:** Features like *timestamp*, *noise_per_ms*, and *jamming_indicator* were removed because they directly or indirectly encode the attack state and would not be reliable indicators in a blind, live production environment. Additionally, features containing exclusively zero or negative one (-1) values were directly removed as they carry no discriminative information.

- 2) **Redundancy and Multicollinearity:** Following established predictive modeling principles for feature selection [11], an upper triangular correlation matrix was computed; any feature pair exceeding a correlation threshold of 0.95 was resolved by removing the less physically interpretable variable (e.g., *alt_ellipsoid* relative to other altitude metrics) to prevent multicollinearity.
- 3) **Perfect Separators:** Several high-precision location metrics and GPS ratios directly provided perfect separation between benign and malicious classes. Because they introduce substantial leakage, direction features (*cog_rad*) and direct location references were discarded.

From the cleaned datasets, two engineered features were derived. The first is *control_std*, which captures control system variability. The second is the total GPS innovation uncertainty (*gps_inn_sum*), computed as the sum of absolute Extended Kalman Filter (EKF) innovation residuals. Mathematically, it is defined as:

$$\text{gps_inn_sum} = \sum_{i=1}^N |x_{\text{inn},i}|, \quad (1)$$

where N is the total number of EKF innovation features and $x_{\text{inn},i}$ represents the i -th innovation residual.

C. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to visualize class distributions (Benign vs. Spoofing/Jamming) to ensure the dataset was manageable without severe imbalance interventions. Box plots and a correlation heatmap (see Fig. 1) were generated to observe relationships between the refined variables and the target classification, indicating that control signals and the derived GPS innovation variables provided the most profound differences across classes.

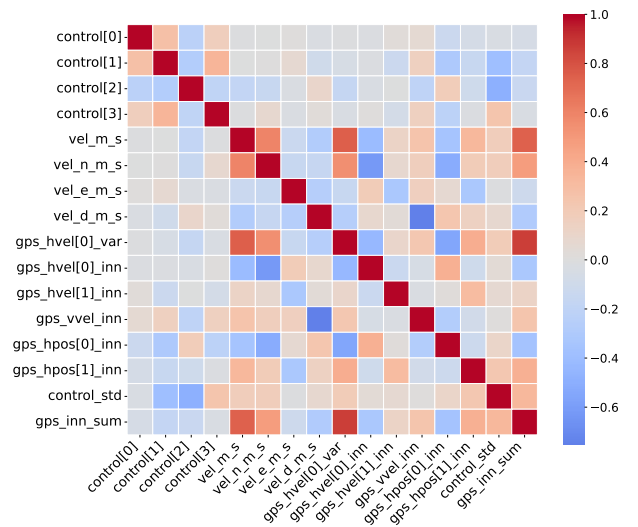


Fig. 1. Feature correlation heatmap identifying the relationships and redundancies among the selected variables post-leakage removal.

TABLE I
OPTIMAL HYPERPARAMETERS (5-FOLD GRIDSEARCHCV)

Model	Parameter	Value
SVM	C	100
	kernel	RBF
	gamma	0.1
RF	n_estimators	100
	max_depth	10
KNN	n_neighbors	3
XGBoost	learning_rate	0.2
	max_depth	5
	n_estimators	200

IV. EXPERIMENTAL SETUP AND ALGORITHMS

A supervised classification methodology was selected, mapping UAV communication patterns into distinct decision boundaries. The dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve the balance of attack types. Furthermore, the numerical input features were standardized so that each feature has a zero mean and unit variance. This ensures that metrics with different units do not arbitrarily dominate distance-based computations. The standardization parameters (mean and variance) were computed exclusively on the training set and subsequently applied to the test set to prevent data leakage during feature normalization.

Four standard ML algorithms and one advanced ensemble method were implemented:

- **Logistic Regression (LR):** Used as a linear baseline for probabilistic classification [12].
- **Support Vector Machine (SVM):** Utilizing a Radial Basis Function (RBF) kernel, SVM was selected for its strength in mapping high-dimensional feature spaces into optimal separating hyperplanes [13].
- **K-Nearest Neighbors (KNN):** Chosen for its non-parametric, instance-based approach capable of capturing non-linear bounds in spatial data [14].
- **Random Forest (RF):** Executed as an ensemble approach capable of handling multi-class problems efficiently, resisting overfitting via bagging, and explicitly calculating feature importance [15].
- **eXtreme Gradient Boosting (XGBoost):** Chosen for its superior capability to model tabular data through a gradient-boosted decision tree architecture [16].

Hyperparameters for the advanced models were tuned using a 5-fold *GridSearchCV*. The optimal hyperparameters determined via this process are provided in Table I.

V. RESULTS AND DISCUSSION

By testing these models on the unseen hold-out dataset, the performance metrics (Accuracy, Precision, Recall, and F1-Score) provide a consistent basis to evaluate their ability to differentiate between Benign Flights, GPS Spoofing, and GPS Jamming.

TABLE II
CLASSIFICATION PERFORMANCE METRICS OF INVESTIGATED MODELS

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	60.33%	60.00%	60.00%	58.00%
SVM	88.26%	88.00%	88.00%	88.00%
KNN	81.90%	82.00%	82.00%	82.00%
Random Forest	91.65%	91.69%	91.65%	91.61%
XGBoost	97.35%	97.00%	97.00%	97.00%

To clearly present the comparative performance of the applied algorithms, the comprehensive results are summarized in Table II.

A. Classification Performance

The models demonstrated varying degrees of success on the leakage-free dataset:

- **Logistic Regression** struggled to isolate the complex relationships, achieving an overall accuracy of 60.33% and an F1-score of 58%. The low performance of Logistic Regression is consistent with the non-linear nature of the feature space, which linear classifiers cannot adequately separate.
- **SVM** achieved a strong baseline after hyperparameter tuning, correctly identifying flight states with an accuracy of 88.26% and a weighted F1-score of 88%.
- **KNN** showed significant improvement in recognizing the spatial clusters of attack states, classifying the test data with 81.90% accuracy and an impressive F1-score of 82%.
- **Random Forest** achieved a high overall accuracy of 91.65%. The specific performance breakdown across classes yielded a weighted precision of 91.69%, recall of 91.65%, and an F1-score of 91.61%.
- **XGBoost** proved to be the most capable model. Due to its advanced gradient boosting architecture, it achieved an exceptional overall accuracy of 97.35%. The specific performance breakdown across classes yielded a weighted precision of 97%, recall of 97%, and an F1-score of 97%, as further detailed in the confusion matrix (see Fig. 2). To validate the real-time applicability of the proposed IDS, the inference latency of the XGBoost classifier was evaluated. On a Google Colaboratory CPU environment (Python 3, runtime version 2026.01), the average prediction time per sample was measured at approximately 0.0146 milliseconds. This extremely low latency ensures that the model can perform real-time threat detection without introducing processing bottlenecks. While these numbers reflect a realistic and mathematically sound reduction compared to the 99% accuracies reported in studies utilizing leakage-prone features, the XGBoost model proves highly effective at accurately identifying threats using strictly realistic metrics.

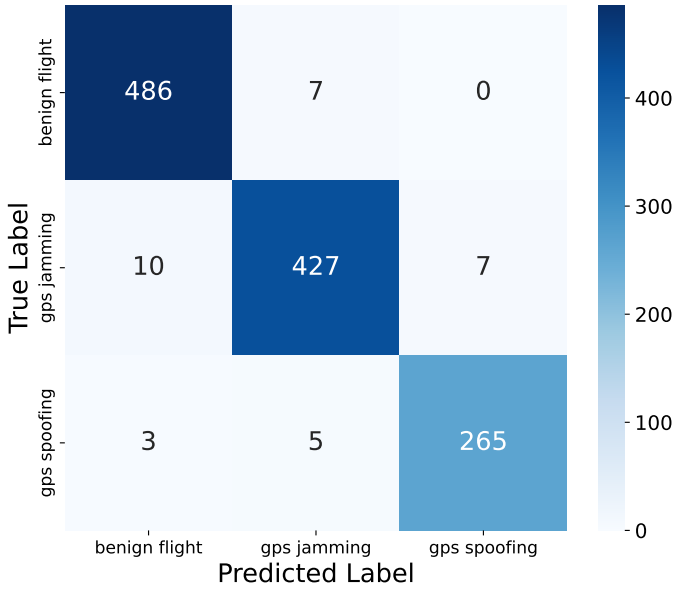


Fig. 2. Confusion matrix of the XGBoost classifier indicating the predicted vs. true labels for Benign, GPS Spoofing, and GPS Jamming flight instances.

B. Feature Importance Analysis

Using the inherent design of the XGBoost model, feature importance scores were aggregated (see Fig. 3). The analysis indicated that specific **control signals**—most notably thrust (*control_3*) and yaw (*control_2*)—along with individual **GPS EKF metrics** (such as *gps_vvel_inn* and *gps_hvel_0_var*) were the most critical determinants for classification. For context within the flight controller architecture [17], the control variables physically map to the UAV’s roll (*control_0*), pitch (*control_1*), yaw (*control_2*), and thrust (*control_3*) commands.

From a cyber-physical perspective, this highlights a clear cause-and-effect relationship during an attack. While the EKF metrics directly capture the sensor discrepancies and signal degradation caused by GPS Spoofing (via innovation) or Jamming (via variance), the control signals reflect the UAV’s mechanical reaction. Because cyber-attacks maliciously manipulate the drone’s perceived position parameters, the onboard flight controller acts aggressively to correct these abrupt deviations. Therefore, observing the variability in the UAV’s control system activity, coupled with EKF confidence metrics, serves as the most robust and realistic indicator of an ongoing cyberattack.

VI. CONCLUSION AND FUTURE WORK

Preprocessing steps to eliminate data leakage proved essential for obtaining reliable IDS performance estimates. By thoroughly cleaning the UAV Attack Dataset of explicit attack identifiers, redundant metrics, and artificially separating location variables, we constructed a more realistic and reliable evaluation benchmark for standard machine learning classifiers.

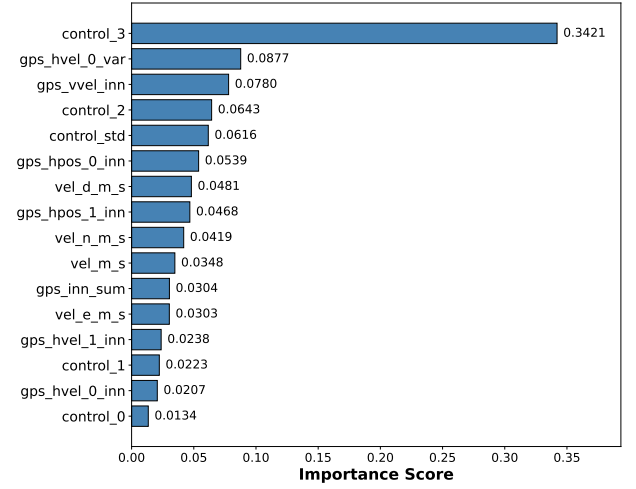


Fig. 3. Feature importance analysis using the XGBoost model, demonstrating the significance of control signals and GPS innovation metrics.

Despite the removal of explicit flags, the engineered feature set allowed the XGBoost classifier to effectively separate the benign data from GPS Spoofing and Jamming attacks, yielding a robust 97.35% accuracy. This indicates that realistic flight telemetry, specifically control signals and GPS innovation features, is fully capable of identifying cybersecurity threats in real-time.

Future work should explore optimizing deeper architectures, such as CNNs or Autoencoders, under the same strict leakage-free conditions. Additionally, investigating the application of federated learning could allow distributed UAV networks to collaboratively develop common threat models without sharing sensitive raw local telemetry data.

DECLARATION

The authors used AI-assisted language tools for grammar refinement and clarity improvement during manuscript preparation. All scientific content, experimental design, and conclusions were developed and verified by the authors.

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